

California Insulated Muni Sleeve

A \$1,000,000, 24-bond California municipal ladder built to keep paying through an AI / technology-sector crash — every pledge verified against the legal record, every price and liquidity figure from the regulatory trade tape, and credit scored on pledge structure with the state's fiscal-stress rosters checked per district.

8.19%

After-tax tax-equivalent yield (assumes CA top combined bracket; scales down with the buyer's marginal rate — e.g. ~×1.5 gross-up at a 35% rate vs ×2.01 here)

4.28%

Yield to worst (par-weighted)

99 /100

AI-insulation score (credit, not price)

87 /100

Independent credit score (pledge-level)

2035–2045

Maturity ladder

25

Positions, \$1.0M par

40%

Liquid-anchor share

~0.2 bp

Amortized annual cost vs 25–35 bp SMA

Buy the pledges that keep paying when tech-driven state revenue collapses.

California's General Fund leans on personal income tax, and the most volatile slice of that is capital-gains realizations on tech equity. A severe AI de-rating would collapse those realizations — **the dot-com bust cut California capital-gains revenue ~71%** and pressured state-General-Fund-backed credits by an estimated 4–5 notches.

What kept paying

Bonds secured by a **statutory property-tax lien** (school general obligation) or by **essential-service utility revenue** were untouched through the dot-com bust. No California school GO has missed a debt-service payment through Loma Prieta, Northridge, or the Camp Fire.

What we exclude entirely

The **state-General-Fund channel** — income-tax-backed state GO and appropriation debt — is the transmission path for an AI shock. This sleeve holds **no state GO or state-appropriation debt**; its only state-linked exposure is two University of California names (8% of par) whose pledge includes some state educational appropriations — scored accordingly at the sleeve's lowest insulation tier.

The yield is not a risk premium for reaching down the quality curve. It is compensation for illiquidity and obscurity inside verified-safe credits — small, unrated-on-the-public-feed, lightly-traded issuers sitting behind a statutory first lien.

WHY THE YIELD EXISTS

An obscurity-and-illiquidity premium, not a credit premium.

Small & unrated on the public feed

EMMA returns no third-party agency rating for most of these issuers — the agencies do not publish them on the public feed. A buyer leaning on agency letters is effectively flying blind, so the names trade cheap to their structure.

Lightly traded

Municipals trade by appointment with no consolidated quote. Thin tapes — some names print a handful of times a month — keep institutional buyers away and widen the spread the seller must give up.

Strong statutory credit underneath

Behind the obscurity sits a **CA Gov. Code §53515 first lien** and an **unlimited ad-valorem levy** that floats up to cover debt service regardless of the economy. The structure is strong; only the visibility and liquidity are weak.

The distinction that matters — liquidity is not credit

Reaching for yield normally means accepting weaker credit. Here the extra yield is paid for **how hard the bond is to find and trade** — a cost borne once, on a hold-to-maturity ladder, against a pledge our own pledge-level read scores as Strong.

Liquidity measures how easily *you* can sell; default risk measures whether the *issuer* pays — and the §53515 levy pays the same whether a bond trades five times a year or five hundred. The most *liquid* municipals have defaulted (Puerto Rico, Detroit GOs); CA school GOs under this lien essentially never have. Thin trading is why these bonds are **cheap** — never why they would be **unsafe**. We are paid for patience, not for risk.

Why one bond costs 79 cents — and another 102.

Worked example — **Turlock USD 3% of 2043 at 79.2 (8.61% after-tax)**. The price looks alarming until you decompose it. None of the three drivers is credit:

1 · Bond math, not distress

These are **3% coupons issued at the 2020–21 rate floor**. With long tax-exempt yields near 4.5%, a 17–19-year 3% coupon *must* price near 78–80 to equate yields. Every 2020-vintage California school 3s-of-'43–'45 looks like this — the price is arithmetic about a coupon, not an opinion about a district.

2 · The de-minimis cliff overshoot

Below the de-minimis threshold, discount accretion is taxed as **ordinary income** — so retail buyers and advisors reflexively avoid these bonds. The market's concession overshoots the actual tax cost: Turlock yields **~4.79%** vs ~4.09% for a comparable par-coupon name — a ~70bp concession against a true tax drag of ~30bp. **The ~40bp net is harvestable only by a top-bracket, hold-to-maturity buyer who does the after-tax math per name.** Buyers who can't, stay away — which is exactly why the concession persists.

3 · Orphan coupon, orphan issuer

The institutional muni standard is a **5% coupon near par**; low-coupon discounts are an unloved structure with no natural constituency. Stack that on small, agency-unrated issuers — Emery USD serves a few hundred students (atop Emeryville's dense commercial tax base) — and the bonds are orphaned twice over, inside the same statutory first-lien pledge as everything else in the book.

The lemons test — structure, not information

If one district yielded 8.6% while its peers sat at 7.5%, someone would know something. Instead **all fifteen discount 3s in the book cluster at 8.0–8.7% after-tax, ordered almost perfectly by maturity** — a whole coupon class being repriced, not name-specific information. And every name-specific channel we can check — fiscal-stress rosters, pledge text, tax status, two-sided tape — came back clean.

The mirror image — why the book also pays 102 for UC Regents at "only" 6.3%

The anchors are the discounts inverted: a **5% UC coupon at 101.7, callable May 2027 at par**, is an **11-month bond wearing a 2035 costume** — 6.32% after-tax to the near-certain call, ~2× a T-bill's after-tax equivalent. It does two jobs the discounts can't: the **front rung of the ladder** (insulated short paper is structurally scarce) and the **deepest liquidity in the book** (~100 trades/yr) — the seat that lets thin names be held without forced selling. Call asymmetry favors the holder (no refund → 5% to 2035 ≈ 9.6% tax-equivalent). Cost of the seat: ~\$700 of forgone yield until the call recycles the cash.

Verified universe → 25 holdings. Every drop is documented.

147

Universe. IBKR scanner pulls 147 California municipal candidates across the insulated sectors (school GO, water/wastewater, electric revenue).

106

Shortlist for primary-record verification. Up to 12 names per maturity year — deliberately mixing discount and premium coupons so the tax math, not the price, decides what wins.

39

Verified clean & tradeable (EMMA per-CUSIP + MSRB tape). Of the 106: **27 were federally taxable** (a naive yield-maximizer would have bought them on a fictitious $\times 2.01$ gross-up they do not earn), **23 failed the liquidity gate** (last trade ≤ 90 days, ≥ 12 trades/yr, ≥ 1 two-sided day), and **17 failed pledge or mark checks** — COPs, lease revenue, judgment-obligation and limited-obligation paper masquerading as school GOs, or stale prints. The marketed label is never trusted; the legal pledge is.

25

Selected on after-tax yield, then verified per district. The 39 survivors are ranked on de-minimis-aware after-tax TEY (one name per issuer, full ladder coverage, fault-zone caps). Each selected district is then resolved (OpenFIGI → CA Dept. of Education) and checked against the state's AB 1200 fiscal-stress rosters: **every K-12 district in the basket is absent from the Negative and Qualified lists** — the stressed districts (SFUSD & Plumas Negative; Oakland & Hayward Qualified) hold no place in this ladder.

Each stage answers a single question against the primary record. The drops are documented per CUSIP — and each rejected category is the kind of mistake a label-trusting screen lets through silently.

How to read the sleeve.

AI-insulation (AI-1 ... AI-5)

Rates each bond's **credit** sensitivity — not its price — to a severe AI/tech crash, through the channel that actually transmits the shock: California capital-gains income tax. Insulation = 100 — exposure.

- AI-1** Insulated — school GO, water rev.
- AI-2** Low — UC general/project rev.
- AI-3** Moderate — tax-increment, hospital
- AI-4** High — local general-fund approp.
- AI-5** Direct — state GF / income-tax GO

Dot-com (2003): AI-1 untouched; AI-5 fell 4–5 notches (CA AA→BBB).

Independent credit (0–100)

Scores the **actual legal pledge**, read from the official statement or governing statute — never the issuer's name or marketed label.

Statutory first-lien school GO	~90 Strong
Essential-service water/sewer	~80 Solid
UC General Revenue Bonds	~82 Solid
UC Limited Project Revenue	~74 Adequate

Lease / COP structures are down-rated for appropriation risk. Any bond whose OS we have not read is flagged **Unverified** — **never treated as clean**. Stands in for the agency letter because EMMA publishes none for most names.

Liquidity (L1 ... L4)

From the **actual MSRB trade tape** — realized round-trip spread, trade frequency, and genuinely two-sided days.

L1 Highly liquid	≤0.25pt · ≥50/yr
L2 Liquid	≤0.40pt · ≥24/yr
L3 Moderate	≤0.75pt · ≥12/yr
L4 Thin	cleared floor

On a hold-to-maturity ladder, spread is paid once on the buy and the bond redeems at par — so liquidity governs **deployment pace and early-exit cost, not income**.

STATISTICAL HIGHLIGHTS

The sleeve at a glance.

8.19%

After-tax TEY (par-wtd)

4.27%

Yield to worst (par-wtd)

99

AI-insulation (par-wtd)

87

Credit (par-wtd)

\$1.0M

Total par · 25 positions

0.358pt

Median same-day spread

42%

Liquid-anchor share (min 40%)

11/25

De-minimis breaches (in TEY)

AI-1 ×23

AI-insulation mix · AI-2 ×2

L1·4 L2·7

Liquidity mix · L3·5 L4·9

Strong ×19

Credit mix · Solid ×5 Adequate
×1

20%

Max single fault-zone (cap 25%)

Selection metric: **after-tax tax-equivalent yield, de-minimis aware**. Discount accretion below the 100 – 0.25×years threshold is taxed as ordinary income; the CA top-bracket gross-up (×2.01) applies only to the tax-exempt component. The after-tax TEY shown already reflects that drag.

How the after-tax yield is actually computed.

The headline metric is an **after-tax tax-equivalent yield** — not a gross coupon. Every discount bond's yield splits into two pieces taxed differently. Worked on **Turlock USD 3% of 2043 at 79.23 (YTW 4.785%)**:

1 · Split the yield by tax character

Coupon (current yield) = $3.00 \div 79.23 = 3.79\%$ — federally **tax-free**.

Accretion to par = YTW – current = $4.785 - 3.79 = 1.00\%$ — **ordinary income**.

Why ordinary? The de-minimis cutoff = $\text{par} - 0.25 \times \text{years} \approx 95.75$ (the bond was issued at 97.80; either way far above 79). Buying below the cutoff makes the entire pull-to-par **ordinary income, not tax-exempt** — the de-minimis rule.

2 · The formula

After-tax cash you keep = coupon (tax-free) + accretion $\times (1 - t)$.

Tax-equivalent yield divides that by $(1 - t)$:

$$\text{TEY} = \text{coupon} \div (1 - t) + \text{accretion}$$

The coupon gets the full gross-up; **the accretion enters at 1.0x — no muni benefit**, because in tax-equivalent terms ordinary income maps 1:1 to a taxable bond. *That 1.0x is the de-minimis penalty, made explicit.*

3 · The after-tax TEY across combined marginal rates

Combined marginal rate t	Gross-up $1/(1-t)$	After-tax TEY	Net yield in pocket
50.25% — the $\times 2.01$ used throughout this deck	2.010	8.61%	4.28%
52.65%	2.112	9.00%	4.26%
54.10% — CA top ordinary (37% fed + 3.8% NIIT + 13.3% CA)	2.179	9.25%	4.25%
55.10% — + CA >\$1M mental-health surcharge	2.227	9.43%	4.24%

The deck uses the **conservative $\times 2.01$ (50.25%)** throughout; a true top-bracket payer is higher, so the figures shown **understate** the benefit by ~50–80 bps. The de-minimis drag itself: a fully-exempt 4.785% would gross up to 9.62% — the 8.61% shown is ~101 bps lower, and that gap is precisely the tax the deep discount carries (the reason these bonds are cheap).

Not tax advice

We are **not tax advisors** and this is **not tax advice**. These calculations are illustrative, assume a single set of current-law California top-bracket rates applied at maturity in 2043–45 (future rates are unknown), and ignore individual circumstances (AMT, residency changes, the annual-accrual election, state of domicile, estate treatment). **Consult a qualified tax professional** before acting; your actual after-tax outcome will differ.

FULL HOLDINGS — ALL 25 POSITIONS

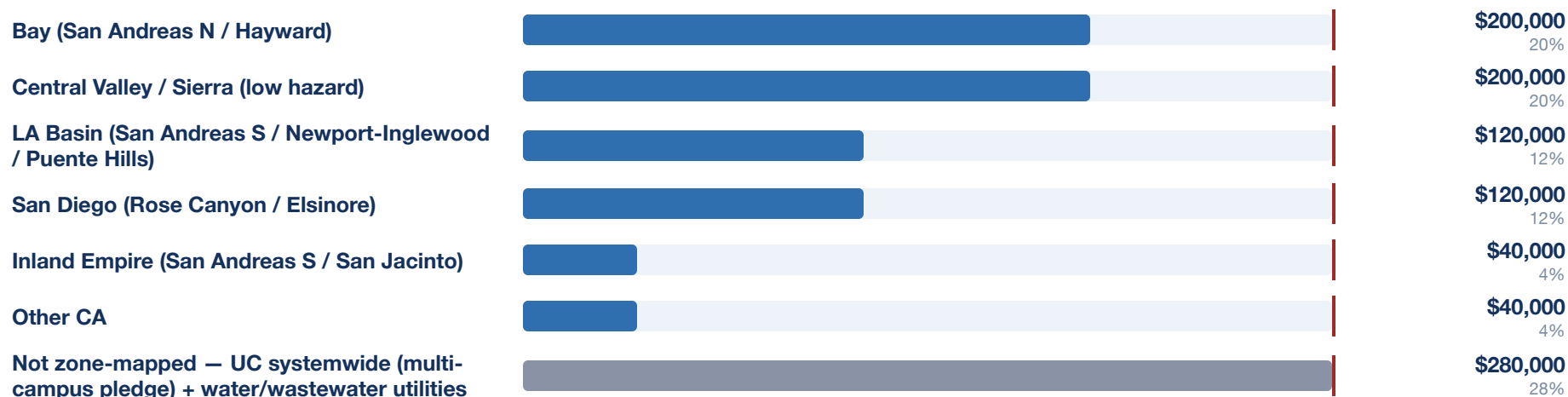
The ladder.

CUSIP	DISTRICT / COUNTY	PLEDGE	SL	CPN	MATURITY	PX	YTW	TEY A-T	LIQ	AI-INSUL	CREDIT	AGENCY (CLASS)	PAR
91412G2F1	UC Regents (LPRB) -	UC project revenue	A	5%	2035-05	101.75	3.14%	6.32%	L1	AI-2 93	74	~Aa3 / AA-	\$40,000
926055KH6	Victor Valley Union High San Bernardino	School GO · SB-222 first lien	A	4%	2037-08	99.99	4.00%	8.04%	L1	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
587619BF3	essential-service water/wastewater -	Essential-service water/sewer rev.	B	4%	2038-10	99.90	4.01%	8.06%	L3	AI-1 100	80	~A1-Aa3 / A+ to AA-	\$40,000
91412HJC8	UC Regents (GRB) -	UC General Revenues	A	4%	2039-05	100.46	3.90%	7.83%	L2	AI-2 87	82	Aa2 / AA+	\$40,000
050002AW4	essential-service water/wastewater -	Essential-service water/sewer rev.	B	5%	2040-05	100.89	3.97%	7.98%	L2	AI-1 100	80	~A1-Aa3 / A+ to AA-	\$40,000
95236EAT2	West County Facilities Financing A -	Essential-service water/sewer rev.	A	4%	2041-06	99.17	4.07%	8.19%	L3	AI-1 100	80	~A1-Aa3 / A+ to AA-	\$40,000
032591TU3	Anaheim Union High Orange	School GO · SB-222 first lien	B	3%	2041-08	83.56 D	4.50%	8.13%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
17132CDL9	Chula Vista Elementary San Diego	School GO · SB-222 first lien	B	4%	2041-08	99.62	4.03%	8.11%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
270198BY9	Earlimart Elementary Tulare	School GO · SB-222 first lien	B	4%	2041-08	99.99	4.00%	8.04%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
358233ED2	Fresno Unified Fresno	School GO · SB-222 first lien	B	3%	2041-08	83.95 D	4.47%	8.08%	L3	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
95330PHH1	W HILLS CMNTY CLG-B -	School GO · SB-222 first lien	A	4%	2041-08	98.93	4.09%	8.23%	L2	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
777387AH4	Roseland Elementary Sonoma	School GO · SB-222 first lien	B	4%	2042-08	97.81	4.19%	8.41%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
817409H32	Sequoia Union High San Mateo	School GO · SB-222 first lien	B	3%	2043-07	82.95 D	4.43%	8.08%	L1	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
607735CA3	Modesto City Elementary Stanislaus	School GO · SB-222 first lien	B	3%	2043-08	81.40 D	4.57%	8.30%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
870462TJ7	Sweetwater Union High San Diego	School GO · SB-222 first lien	B	4%	2043-08	98.57	4.12%	8.27%	L2	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
900211CP6	Turlock Unified Stanislaus	School GO · SB-222 lien (SFID)	B	3%	2043-08	79.23 D	4.79%	8.61%	L2	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
13049WAX3	essential-service water/wastewater -	Essential-service water/sewer rev.	B	4%	2044-06	98.77	4.10%	8.24%	L4	AI-1 100	80	~A1-Aa3 / A+ to AA-	\$40,000
168520PA6	Chico Unified Butte	School GO · SB-222 first lien	B	3%	2044-08	79.04 D	4.73%	8.56%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
171314LB1	Chula Vista Elementary San Diego	School GO · SB-222 first lien	B	4%	2044-08	100.06	3.99%	8.02%	L2	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
547541LJ9	Lowell Joint Orange	School GO · SB-222 first lien	B	3%	2044-08	78.12 D	4.82%	8.70%	L3	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
012104QQ1	Albany City Unified Alameda	School GO · SB-222 first lien	B	3%	2045-08	77.20 D	4.84%	8.76%	L3	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
079113DY9	Bellevue Union Sonoma	School GO · SB-222 first lien	B	3%	2045-08	80.66 D	4.51%	8.27%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
223093TK1	Covina-Valley Unified Los Angeles	School GO · SB-222 first lien	B	3%	2045-08	78.67 D	4.69%	8.54%	L4	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
291119KP9	Emery Unified Alameda	School GO · SB-222 first lien	B	3%	2045-08	77.94 D	4.77%	8.66%	L2	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000
796472AS7	San Benito High San Benito	School GO · SB-222 first lien	B	4%	2045-08	98.75	4.09%	8.23%	L1	AI-1 99	90	~Aa2-Aa3 / AA- to AA	\$40,000

D = de-minimis breach (accretion taxed as ordinary income; reflected in after-tax TEY). **SI** = sleeve (A liquid-anchor / B yield-core). **Px** = EMMA last-trade. Agency (class) is a security-class benchmark, not an issue-specific rating — EMMA publishes no third-party rating for most of these issuers.

Earthquake: cap the zones, trust the levy.

Par by fault zone — cap 25% per zone (dashed line)



The zone bars cover the 72% of par with a single mappable district location. The remaining 28% is geographically diffuse (UC's pledge spans ten campuses statewide) or utility-revenue-backed, but is not zone-scored — the 20%-max-zone figure applies to the mapped portion only.

Why the levy survives a quake

The unlimited ad-valorem levy **floats up on surviving assessed value** to cover debt service; Prop 13 keeps assessed value well below market, a buffer against decline; county **Teeter** plans advance delinquent collections to the district. No CA school GO defaulted through Loma Prieta, Northridge, or the Camp Fire. The 25% per-zone cap reflects that **San Andreas North (1906) and South (1857) rupture independently**.

What could still hurt

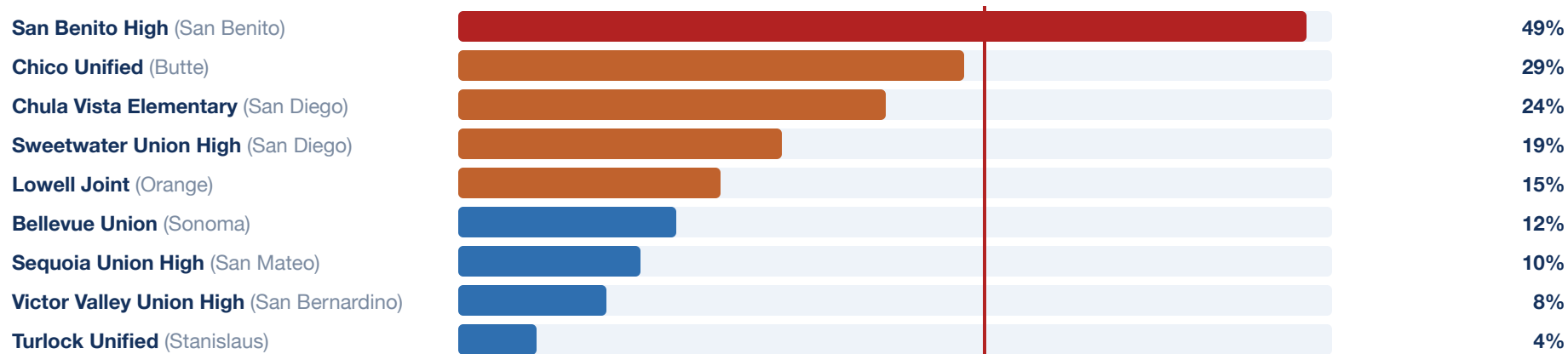
Rate shock: all munis can fall together — AI-insulation rates credit, not price. **Fill slippage:** thin names give back some yield in spread on the buy. **Community-college names:** AB 1200 certification does not apply (CCCO monitors); their fiscal status is **unverified, not clean**. **Concentration:** Bay + LA Basin together are the bulk of the ladder.

AB 1200 fiscal-certification gate

Every K-12 district in the basket is **absent from the state's Negative and Qualified rosters** (FY24-25 interims). The publicly stressed districts — **SFUSD & Plumas (Negative), Oakland & Hayward (Qualified)** — hold no place in this ladder. Community-college districts are outside this regime: treated as unverified, not clean.

Wildfire: the slower peril the earthquake hedge concentrates.

Share of district building value in FEMA high/very-high wildfire tracts — red line = 30% cap threshold



Scale: full bar = 50%. Names above 30% are capped at 15% of the book; the basket holds **one** (San Benito, insured). UC, water-utility and one reorganized district are not district-mappable and are recorded as **unverified, not clean**.

The channel is insurance, not flame

No CA school GO defaulted through Loma Prieta, Northridge, or the Camp Fire — Paradise's bonds paid while the town burned. The live risk is slower: **insurer withdrawal** makes property unmarketable, which drags the assessed-value base the levy rides on over years, with no covenant trigger. That benign historical record predates this channel — which is exactly why we measure exposure instead of citing history.

Why the two perils share a page

Steering the book away from fault zones steers it toward the hills — **the first draft of this portfolio concentrated 21% of names in high-fire districts because of its seismic discipline**. Three were swapped out at zero yield cost once fire was scored. Earthquake is capped per fault zone (ruptures are local: 1906 vs 1857); fire is capped in aggregate (sieges are statewide: 2017, 2020). One peril per slide, one joint constraint set per book.

DEPLOYMENT & COSTS

Invested from day one; the rest accumulated patiently.

Phase 1 — deploy now

~\$400,000 (40%)

The liquid anchor plus already-active yield-core names — buildable on demand at tape-tight spreads. You earn the full after-tax TEY from day one.

Phase 2 — accumulate over weeks

~\$600,000 (15 names)

The thinner yield-core names, worked opportunistically on offerings and bids-wanted, funded by trimming the anchor as they fill. Worst case you simply hold the liquid anchor longer — it still out-yields cash.

Cost stack

~\$215

Total IBKR muni commission — ~2.2 bps one-time, ~0.2 bp/yr amortized over the ladder.

0.1–0.5pt

The real cost — dealer markup on thin names, paid once on the buy. Phase 1 avoids most of it at tape-tight spreads.

The price of patience — Phase 2 illiquidity, measured from its own tape

+22 bps/yr

Illiquidity premium earned: patient names yield 8.28% vs 8.05% liquid (~\$1,340/yr).

≈3 bps/yr

Entry cost: ~0.52pt mean half-spread, ~\$3.1k one-time, amortized to maturity — a ~7:1 earn-to-cost ratio.

4–8 wks

Build window working all names in parallel; 12/15 names have printed ≥\$40k blocks.

~2.5%

Forced-exit cost under stress (≈\$14.8k ≈ one year's premium) — the bargain assumes hold-to-maturity.

Against a municipal SMA at **25–35 bps/yr**, the all-in direct-ownership cost is roughly an order of magnitude lower — and the bonds redeem at par, so commission, markup and the patience premium are paid once, not annually.

HONEST CAVEATS — WHAT COULD MAKE THIS WRONG

Read these before the headline yield.

- ! **Marks are EMMA last-trade prices.** Municipals trade by appointment, so a given bond's last print may be days old even for liquid names. The sleeve's value is an appointment-market estimate, not a continuous quote.
- ! **Thin names give back some yield in spread on fill.** The realized after-tax TEY on the yield-core depends on execution; the dealer markup (0.1–0.5pt) on the buy is a real reduction to the headline figure for the L4 names.
- ! **AI-insulation rates credit, not price.** It is a statement about which pledges keep paying in a tech-driven state-revenue collapse — not a forecast of mark-to-market in a pure interest-rate shock, where all munis can fall together.
- ! **The agency column is a class benchmark, not an issue rating.** EMMA carries no issue-level third-party rating for most of these names; the column shows the *typical* rating for each security class for context only. Issue-level ratings can be pulled from a paid feed on request.
- ! **Community-college districts are unverified-not-clean.** AB 1200 interim certification does not apply to them (the Chancellor's Office monitors separately); we have not independently cleared their fiscal status, and we do not treat that absence as a clean bill.
- ! **Eleven of twenty-four bonds breach the de-minimis threshold.** Their discount accretion is taxed as ordinary income. The after-tax TEY already reflects this drag, but it makes the tax outcome sensitive to the buyer's actual bracket and holding behavior.
- ! **This is not investment advice.** It is verification-driven research for analyst review. No alpha or backtested outperformance is claimed — the pitch is verified safety, tax-efficient yield, and low cost, honestly stated.